Final Internship Project Report

On

AI Career Pulse: Decoding the Job Market (2025-2026)

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1. Introduction

1.1. Project Overview

This project, AI Career Pulse: Decoding the Job Market (2025–2026), analyzes the global and Indian AI hiring landscape. As AI transforms fields from healthcare to finance, this study uses Tableau Desktop Public Edition and Flask web hosting to turn raw job market data into actionable visual insights for candidates and recruiters.

1.2. Objectives

- **Map Salary Levels:** Track average, minimum, and maximum salaries across various AI job roles.
- **Analyze Hiring Personas:** Provide specific market paths for Freshers, Mid-Level professionals, and Senior leaders.
- **Compare Company Scale & Industries:** Analyze salary spending across Startups, Mid-size firms, and Enterprise companies, as well as industry sectors like Government, Retail, Healthcare, and Finance.
- **Interactive Web Hosting:** Integrate Tableau Dashboards into a lightweight Flask web application for browser access.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Job seekers at different career stages face unique obstacles:

- Freshers struggle with a lack of practical experience and uncertainty about entry-level roles.
- Mid-Level Professionals face career stagnation and need to transition into AI without losing domain expertise.
- Senior Professionals face automation risks and must shift toward AI governance, strategy, and architecture.

2.2. User Persona Scenarios & Proposed Solutions

- **Scenario 1: Fresher Perspective:** Focuses on entry-level salary bands ($100k) and industries with active entry hiring (e.g., Government, Healthcare) to build foundational skills.
- **Scenario 2: Mid-Level Professional Perspective:** Focuses on transferable skills, transition roles (e.g., Data Engineer, ML Engineer), and mid-tier salary bands ($100k–$150k).

- **Scenario 3: Senior Professional Perspective:** Focuses on high-paying specialized roles (e.g., Multimodal AI, Infrastructure) and enterprise demand ($261k max salaries) to pivot into AI leadership and governance.

2.3. Initial Project Planning

- **Phase 1 (Data Prep):** Cleaning job posting attributes, validating numerical ranges, and creating standard metrics.
- **Phase 2 (Visualization):** Designing individual visual components in Tableau Desktop.
- **Phase 3 (Dashboarding & Storytelling):** Assembling interactive dashboards and narrative story points.
- **Phase 4 (Documentation):** Authoring GitHub documentation, generating the project report, and recording a demonstration video.

3. Data Collection and Preprocessing Phase

3.1. Data Collection & Identified Attributes

- **Dataset: ai_jobs_market_2025_2026**
- **Core Attributes:** Job Title, Job Category, Company Size, Industry, Country, Education Required, Annual Salary Usd, Years Of Experience, Posting Year, Is LLM Role, and Remote Work.

3.2. Data Quality & Cleaning

- **Data Integrity:** All salary metrics were standardized to USD integers.
- **Calculated Metrics:** Built aggregated fields including AVG(Annual Salary Usd), AVG(Salary Max Usd), AVG(Salary Min Usd), MIN(Years Of Experience), and CNTD(Job Id).
- **Primary Slicing Filters:** Focused on Country: India and Education Required: Associate's for baseline market comparisons.

4. Data Visualization

4.1. Visualizations Created

The project includes the following 7 core visualizations (out of 9 total worksheets):

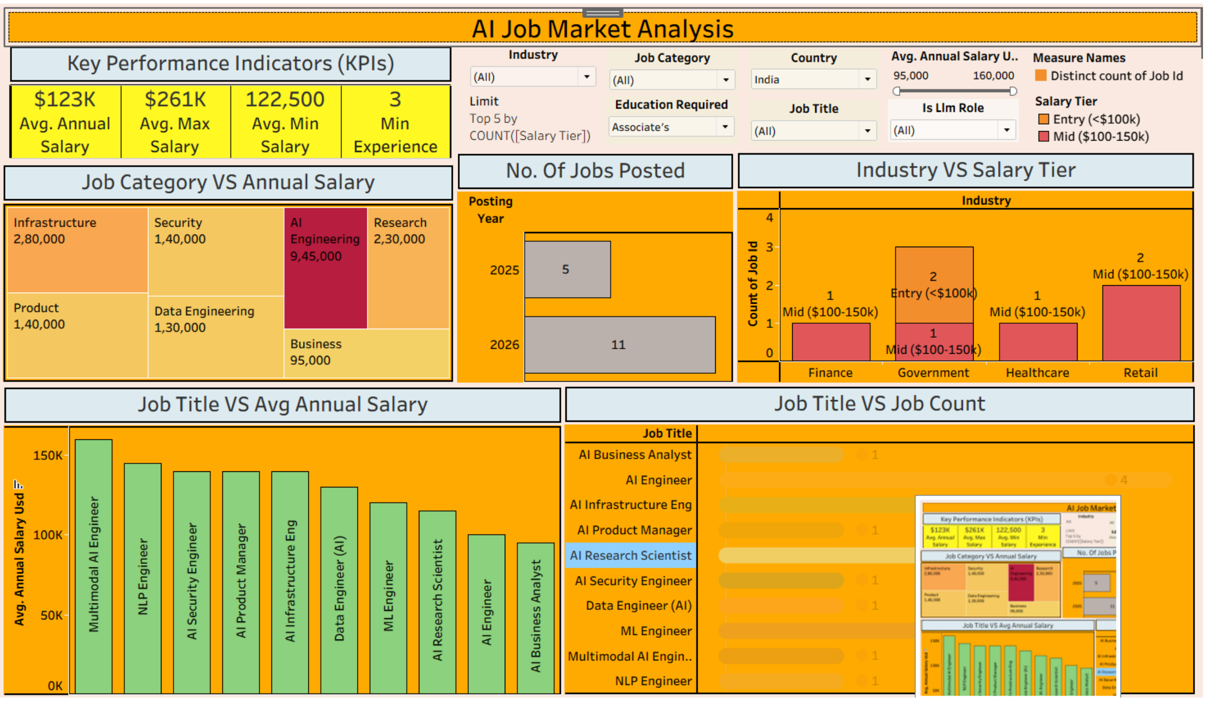
1. **Job Category VS Annual Salary (Treemap):** Shows overall budget allocations (AI Engineering leads at $945,000, Infrastructure at $280,000, and Research at $230,000).
2. **Job Title VS Avg Annual Salary (Bar Chart):** Displays specialized roles (Multimodal AI Engineer highest at ~$160,000, followed by NLP Engineer at ~$145,000).

3. **Company Size VS Annual Salary (Horizontal Bar):** Compares budget sizes (Enterprise at ~$730,000, Startups aggressively bidding at ~$680,000).
4. **Industry VS Salary Tier (Stacked Bar Chart):** Shows hiring across sectors (Government leads with 3 postings in Entry/Mid tiers; Retail, Finance, and Healthcare sit in the Mid $100k–$150k range).
5. **Job Title VS Job Count (Multi-color Bar Chart):** Identifies hiring volume (AI Engineer leads with 4 openings).
6. **No. Of Jobs Posted (Year-over-Year):** Tracks growth from 5 postings in 2025 to 11 postings in 2026 (a 120% YoY increase).
7. **Key Performance Indicators (KPIs)**

5. Dashboard & Web Integration

5.1. Key Performance Indicators (KPIs)

- **Avg. Annual Salary: $123K**
- **Avg. Max Salary: $261K**
- **Avg. Min Salary: $122.5K**
- **Min Experience Required: 3 Years**



5.2. Web Hosting Architecture

- The Tableau dashboard and story points are embedded into a Flask Python application, allowing seamless web rendering via app.py, templates/index.html, and static/style.css.

6. Report

6.1. Story Design File

The Tableau Story Workbook (Tableau Public - Final internship project story) is organized chronologically to guide users through the analytical narrative:

1. **Executive Summary KPIs:** Establishing baseline compensation metrics.
2. **Category & Role Deep-Dive:** Moving from broad Treemap categories to granular job title averages.
3. **Market Dynamics:** Evaluating company size spending and industry salary tier stacks.
4. **Growth Trajectory:** Demonstrating year-over-year expansion from 2025 to 2026.

7. Performance Testing & Interactive Filters

- **Applied Filters:** Integrated interactive filters for Job Title, Experience Level, Industry (Top 5), Remote Work, Is LLM Role, and Posting Year.
- **Performance:** Client-side dashboard interactions run smoothly without query delay.

8. Conclusion & Key Observations

- **High Volume vs. High Salary:** Generalist AI Engineers drive the highest hiring volume, while niche technical roles (Multimodal AI, NLP) command top individual salary rates.
- **Startup Competitiveness:** Startups (1–50 employees) offer salary budgets (~$680K) nearly equal to large Enterprises (~$730K) to secure top talent.
- **Market Acceleration:** Total job postings doubled between 2025 and 2026 (120% growth).

9. Future Scope

In the future, this project can be expanded and improved in the following ways:

- **Live Data Updates:** Instead of using static data, connect the dashboard directly to live job websites (like LinkedIn or Indeed) to automatically show new job postings as they appear.
- **Predictive Salary Model:** Use Machine Learning in Python to build a smart tool that predicts a job seeker's expected salary based on their experience, skills, and location.
- **Global Market Expansion:** Add deeper analysis for more countries (like the US, UK, Canada, and Germany) to compare how AI salaries in India match up against global trends.

- **Skill Matching Tool:** Create a feature where users can select their current technical skills (such as Python, SQL, or PyTorch) to get personalized AI career recommendations and see missing skills they should learn.
- **Enhanced Mobile View:** Optimize the Flask web application and Tableau layouts so that the dashboards are easy to view and interact with on mobile phones and tablets.

10. Appendix & Live Project Links

- **GitHub Repository:** AI-Career-Pulse-Decoding-the-Job-Market
- **Live Tableau Dashboard:** Dashboard 1 on Tableau Public
- **Live Tableau Story:** Story 1 on Tableau Public
- **Project demo link:** https://drive.google.com/file/d/1AhjyGxxGmExk16MxuLoX95EPAXIpRsKx/view?usp=drive_link